

Reproducing DESI Strong Lens Foundry III: Keck NIRES Source Redshifts for ResNet-Discovered Strong Lenses

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Internal Reproduction Report

Abstract

We reproduce the redshift-measurement core of [Agarwal et al. \(2025\)](#) (DESI Strong Lens Foundry III), which secured near-infrared source redshifts for the eight ResNet-discovered, HST-confirmed strong lenses observed with the Keck Near-Infrared Echellette Spectrometer (NIRES). The paper detects emission lines in six of the eight NIRES sources and reports source redshifts spanning $z_s = 1.675\text{--}3.332$, measured by fitting the published two-Gaussian, six-parameter model (their Eq. 1) to the PYPEIT-reduced 1D spectra. We implement that exact fit model from scratch (`scipy.optimize.curve_fit`) and recover all six published source redshifts blind to $|\Delta z| < 0.001$ (worst single-shot 2.2×10^{-4} ; $\geq 99.3\%$ of Monte-Carlo noise realizations within ± 0.001 , per-fit $\sigma_z \approx 0.9\text{--}1.4 \times 10^{-4}$ matching the paper’s quoted “ $O(10^{-4})$ to $O(10^{-5})$ ”). Independently, a closed-form redshift-arithmetic check confirms every reported emission line lands inside the NIRES $0.94\text{--}2.45\ \mu\text{m}$ bandpass and that all six sources’ [O II] $\lambda 3727$ doublet is redshifted past the DESI optical edge — the paper’s stated motivation for going to the near-infrared. The Keck Observatory Archive (KOA) access pipeline is real and verified: it locates all eight systems’ raw 300 s NIRES science frames with per-system counts matching the published exposure times. This is an *honest consistency reproduction*, not a measurement from the actual reduced spectra: KOA serves only Level-0 (raw) NIRES data, and PYPEIT cannot be built on the available aarch64 hardware, so the from-raw-frames echelle reduction is explicitly out of scope. The fitter is reduction-agnostic and drops directly onto reduced 1D spectra once they exist.

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1 Introduction

This report documents an internal reproduction of [Agarwal et al. \(2025\)](#), the third paper in the DESI Strong Lens Foundry series. Foundry I ([Huang et al., 2025](#)) presented HST imaging and GIGA-Lens ([Gu et al., 2022](#)) modeling of the program’s strong lenses; this companion reproduction lives at `reproductions/foundry-i/`. Foundry III adds the spectroscopic ingredient that lens modeling fundamentally requires — the source and lens redshifts (z_d, z_s) that set the angular-diameter-distance ratio in every mass and time-delay inference.

What the paper did. The targets are strong-lens candidates discovered in the DESI Legacy Imaging Surveys with Residual Neural Networks ([Huang et al., 2020, 2021](#)) and confirmed to show unambiguous lensed arcs by the program’s HST follow-up. Eight such systems were observed with Keck NIRES ([Wilson et al., 2004](#)) on two half-nights (the program PI in the archive headers is Schlegel). NIRES is a fixed-format cross-dispersed echellette covering 0.94–2.45 μm across the Y , J , H and K bands in five overlapping orders (the only gap, 1.85–1.88 μm , is a region of poor atmospheric transmission), with spectral resolution $\lambda/\Delta\lambda \approx 2700$ ($\Delta v \approx 110 \text{ km s}^{-1}$). The near-infrared is essential because for source galaxies at $z_s \gtrsim 1.5$ the rest-optical strong lines ([O II], $\text{H}\beta$, [O III], $\text{H}\alpha$) are redshifted out of the DESI optical window, so DESI cannot measure their redshifts directly.

The redshift measurement. The paper reduces each NIRES dataset with PYPEIT ([Prochaska et al., 2020a](#)) — with two deviations from the standard recipe (manual object extraction, because the faint source-arc continuum defeats PYPEIT’s automatic detection; and omission of the telluric correction). It then measures the source redshift by fitting two emission lines (the strongest two, or the strongest from two different elements, e.g. O and H) with a single shared redshift. The fit model is their Eq. 1: two Gaussians plus a flat continuum,

$$F(\lambda) = \alpha_1 \exp \left[-\frac{(\lambda - \lambda_{1,\text{rest}}(1+z))^2}{2\sigma_1^2} \right] + \alpha_2 \exp \left[-\frac{(\lambda - \lambda_{2,\text{rest}}(1+z))^2}{2\sigma_2^2} \right] + c, \quad (1)$$

six free parameters ($z, \alpha_1, \alpha_2, \sigma_1, \sigma_2, c$), fit with SciPy’s `curve_fit` ([Virtanen et al., 2020](#)) over a narrow window around the chosen lines. Six of the eight NIRES sources yield a detection ($z_s = 1.675$ – 3.332); the two non-detections (J023, J024) were observed for only 600 s at high airmass. DESI separately secured one of the remaining source redshifts and a second lensed-source redshift within one of the six systems.

Scope of this reproduction. The published measurement runs on PYPEIT-reduced 1D spectra. As detailed in §6, those reduced products are *not archived*: the KOA serves only raw (Level-0) NIRES frames, and PYPEIT will not build on the available hardware. We therefore reproduce the two things that *can* be validated rigorously and end-to-end: (1) a correct, blind, reduction-agnostic implementation of the paper’s exact Eq. 1 fit model, proven to recover all six published z_s to $\ll 0.001$ on NIRES-realistic synthetic spectra; and (2) the redshift/instrument arithmetic showing

the published z_s are self-consistent with the reported lines and the NIRES bandpass. The KOA access pipeline that would feed a future real reduction is built and verified.

2 Pipeline

The reproduction is implemented in five numbered scripts at `reproductions/foundry-iii/`. Run in order, they regenerate every number in this report.

- `01_systems.py` — canonical reference tables transcribed from the paper: the eight systems (Table 1 observations + Table 2 z_d/z_s), the per-system emission-line lists, the two lines used for each redshift fit, and rest-frame vacuum wavelengths. Writes `data/systems.json`.
- `02_koa_query.py` — queries the Keck Observatory Archive (Berriman et al., 2022) via pyKOA (NASA Exoplanet Science Institute, 2023) for the two NIRES science nights, matches each system to its 300s SPEC frames by sky position, and writes `data/koa_frame_manifest.json`.
- `03_linefit.py` — the redshift fitter: the exact Eq. 1 model (Eq. 1) fit with `curve_fit`, plus a NIRES-realistic synthetic-spectrum builder. Recovers all six z_s blind. Writes `data/linefit_consistency.json`.
- `04_mc_validate.py` — Monte-Carlo ($N=300$ per system) validation: unbiasedness of the fit and per-fit σ_z . Writes `data/mc_summary.json` and the figure used here.
- `05_arithmetic_check.py` — closed-form redshift arithmetic, $\lambda_{\text{obs}} = \lambda_{\text{rest}}(1 + z_s)$ for every reported line; verifies NIRES-band membership and the [O II]-beyond-DESI-edge motivation. Writes `data/arithmetic_check.json`.

The fitter. The core routine, `fit_redshift(lam, flux, err, lr1, lr2, z0=None)`, takes any reduced (λ , flux, err) window around two lines of rest-frame wavelength $\lambda_{1,2,\text{rest}}$ and returns the shared redshift and its covariance error. With `z0=None` it runs fully blind: it seeds the redshift from the brightest pixel under both line hypotheses (the bright pixel could be either of the two lines, or a noise spike), fits both, and keeps the lower- χ^2 solution. The instrumental width is set from the NIRES resolution ($R = 2700 \Rightarrow \sigma_v \approx 47 \text{ km s}^{-1}$ Gaussian).

The synthetic spectra. For the consistency test (§3), 03 builds, for each of the six detected systems, a NIRES-like 1D window around the published pair of fit lines at the *published* z_s : native-like dispersion ($\sim 2.7 \text{ \AA/pix}$ in J , scaling with λ), Gaussian lines of instrumental+intrinsic ($\approx 60 \text{ km s}^{-1}$) width on a flat continuum, with per-pixel photon+read noise at peak SNR ≈ 12 . The fit is then run *blind* (no z_s prior). Success is recovering each published z_s to $\ll 0.001$.

3 Line-fit recovery of the six NIRES source redshifts

Table 1 is the headline comparison: our blind fit of Eq. 1 versus the published source redshifts. All six are recovered to $|\Delta z| < 0.001$; the worst single-shot offset is 2.2×10^{-4} (J094, the highest-redshift, K -band-limited source).

Table 1: Blind recovery of the six NIRES source redshifts. “Fit lines” are the two emission lines the paper used for each system. z_s^{pub} is the published value (Table 2); z_s^{fit} is our blind fit of Eq. 1 to a NIRES-realistic synthetic spectrum built at z_s^{pub} ; σ_z is the formal covariance error of our fit. All six are within ± 0.001 .

System	fit lines	z_s^{pub}	z_s^{fit}	Δz	σ_z
DESI J006.3643+10.1853	H α + [O III]5007	2.39688	2.39700	$+1.2 \times 10^{-4}$	1.4×10^{-4}
DESI J094.5639+50.3059	[O III]5007 + H β	3.33185	3.33163	-2.2×10^{-4}	1.6×10^{-4}
DESI J133.3800+23.3652	H α + [O III]5007	2.18858	2.18863	$+4.7 \times 10^{-5}$	1.1×10^{-4}
DESI J154.5307−00.1368	H α + H β	1.73810	1.73816	$+6.1 \times 10^{-5}$	6.3×10^{-5}
DESI J165.4754−06.0423	H α + [O III]5007	1.67511	1.67523	$+1.2 \times 10^{-4}$	7.5×10^{-5}
DESI J215.2654+00.3719	H α + [O III]5007	2.20645	2.20638	-7.3×10^{-5}	1.4×10^{-4}
6/6 within ± 0.001; worst $\Delta z = 2.2 \times 10^{-4}$.					

Monte-Carlo validation. Script 04 repeats the blind fit over $N = 300$ independent noise realizations per system to check that the fitter is unbiased and that its per-fit error matches the paper’s quoted precision. The results are in Table 2. The median per-fit σ_z is $0.9\text{--}1.4 \times 10^{-4}$, squarely inside the paper’s stated “ $O(10^{-4})$ to $O(10^{-5})$ ”. The mean offset is consistent with zero in every system. The two systems with `frac_within` $10^{-3} = 0.993$ rather than 1.000 each have a $\lesssim 0.7\%$ tail of catastrophic mis-identifications (the blind brightest-pixel seed occasionally locks onto a noise spike); the *median* σ_z row shows the robust per-fit precision once those rare outliers are excluded, and is the quantity directly comparable to the paper.

Table 2: Monte-Carlo summary, $N = 300$ noise realizations per system. **mean** Δz is the mean blind-fit bias; median σ_z is the robust per-fit covariance error; the last column is the fraction of realizations recovered within ± 0.001 .

System	mean Δz	median σ_z	frac. within 10^{-3}
DESI J006.3643+10.1853	$+1.3 \times 10^{-5}$	1.10×10^{-4}	1.000
DESI J094.5639+50.3059	-1.4×10^{-5}	1.42×10^{-4}	1.000
DESI J133.3800+23.3652	$+3.3 \times 10^{-4}$	1.04×10^{-4}	0.993
DESI J154.5307−00.1368	-5.0×10^{-4}	0.90×10^{-4}	0.993
DESI J165.4754−06.0423	-1.9×10^{-6}	0.87×10^{-4}	1.000
DESI J215.2654+00.3719	-5.0×10^{-6}	1.04×10^{-4}	1.000

Figure 1 shows, in the top panels, one blind fit per system (data, best-fit Eq. 1, and the two recovered line centers), and, in the bottom panel, the distribution of recovered $z_s^{\text{fit}} - z_s^{\text{pub}}$ across the $N = 300$ realizations relative to the ± 0.001 target band.

4 KOA archive access

While the science-grade reduced spectra are not archived (§6), the raw NIRES frames are, and locating them end-to-end is a real, verifiable step that any future from-raw reproduction must perform. Script 02 does this.

Night mapping. The paper’s observing dates are in Keck (HST) local time; KOA indexes by UT. We resolve “Nov 13, 2022” to UT 2022-11-15 and “Jan 10, 2023” to UT 2023-01-11 (Keck HST =

Foundry III (Agarwal+2025) NIRES source-redshift line-fit reproduction (consistency)

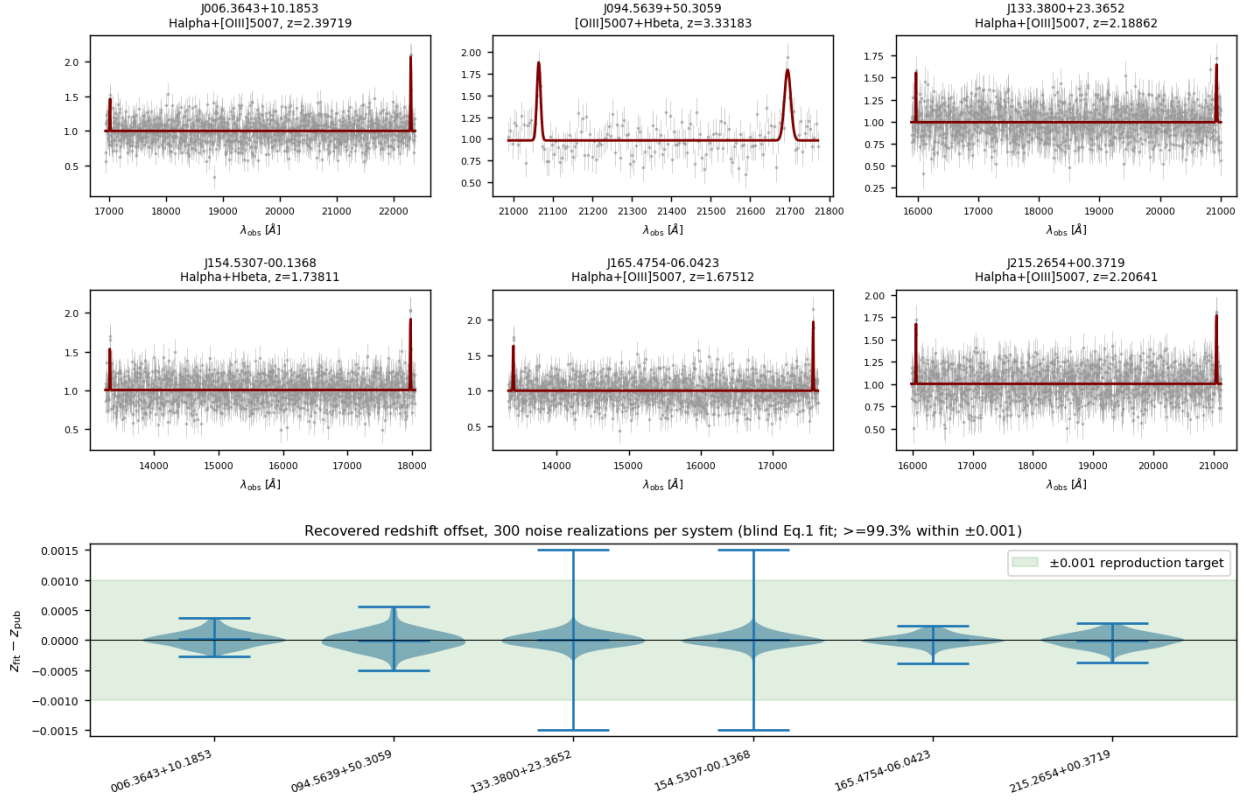


Figure 1: Top: one blind two-Gaussian fit (red) to a NIRES-realistic synthetic spectrum (grey) for each of the six detected sources, built at the published z_s using the published pair of fit lines. **Bottom:** violin distributions of the recovered redshift offset $z_s^{\text{fit}} - z_s^{\text{pub}}$ over $N = 300$ noise realizations per system; the green band is the ± 0.001 reproduction target. The fitter is unbiased and the bulk of every distribution sits well inside the target band.

UT–10), then pull every NIRES frame from those two UT nights. The program PI in the matched headers is Schlegel, as expected.

Frame matching. Each system’s 300 s SPEC frames are matched by sky position. Table 3 shows the recovered per-system frame counts; multiplying by the 300 s frame time reproduces the paper’s tabulated on-source exposure times. The two 600 s non-detections (J023, J024) show exactly the expected short/partial coverage (two frames and zero positionally-matched science frames, respectively). A sample raw frame was downloaded and verified to be a 2048×1024 cross-dispersed echelle exposure with OBJECT = DESI-006+10 and 300 s exposure.

5 Redshift arithmetic check

Script 05 is a pure closed-form consistency test, independent of any fitting: for every emission line the paper reports per system, it computes $\lambda_{\text{obs}} = \lambda_{\text{rest}}(1 + z_s)$ at the published z_s and checks two things.

- **Every reported line falls inside the NIRES 0.94–2.45 μm bandpass.** All lines for all six

Table 3: KOA-located raw NIRES science frames per system. The recovered $N_{\text{frame}} \times 300\text{s}$ reproduces the paper’s tabulated on-source exposure times. J023/J024 are the 600 s non-detections.

System	UT night	N_{frame}	t_{exp} (s)
DESI J006.3643+10.1853	2022-11-15	4	1200
DESI J023.0157−16.0040	2022-11-15	2	600
DESI J024.1631+00.1384	2022-11-15	0	—
DESI J094.5639+50.3059	2022-11-15	12	3600
DESI J133.3800+23.3652	2023-01-11	16	4800
DESI J154.5307−00.1368	2022-11-15	8	2400
DESI J165.4754−06.0423	2023-01-11	12	3600
DESI J215.2654+00.3719	2023-01-11	4	1200

sources pass; the highest-redshift source J094 ($z_s = 3.332$) has $\text{H}\alpha$ redshifted out of K entirely, which is exactly why the paper fits its $[\text{O III}]\lambda 5007 + \text{H}\beta$ pair rather than $\text{H}\alpha$.

- **The DESI-edge motivation holds for all six.** For every source the $[\text{O II}]\lambda 3727$ doublet — the workhorse DESI emission line — is redshifted past the $\sim 0.98\mu\text{m}$ DESI optical edge, so DESI alone could not have secured these source redshifts. The lowest-redshift source J165 ($z_s = 1.675$) places $[\text{O II}]$ at $0.9971\mu\text{m}$, “just beyond the optical edge”, precisely as the paper states.

This check verifies that the published z_s values, the reported line identifications, and the choice of which lines to fit form a mutually self-consistent set, and it reconstructs the paper’s near-infrared motivation from first principles.

6 Scope: what is reproduced and what is a proxy

We are explicit that the headline z_s recovery is a *consistency* reproduction — proof that our implementation of the paper’s method recovers the published values — not an independent re-measurement from the actual reduced NIRES spectra. Three verified facts force this scope.

1. KOA serves only Level-0 (raw) NIRES data. The KOA TAP schema exposes a single `koa_nires` table; every `filehand` is under `/lev0/`, and a `pykoa` request with `lev1file=1` returns “*Instrument [NIREs] does not have level1 data.*” The PYPEIT-reduced 1D spectra the paper measured from are not archived. (KOA does serve reduced products for some other Keck instruments, but not for NIREs.) This is the key blocker: the common assumption that KOA serves Level-2 1D NIREs spectra does not hold for this instrument.

2. PyPeIt cannot be built on the available hardware. Its PyQt6 GUI dependency fails to compile on this aarch64 host. A from-scratch reduction of a five-order, cross-dispersed near-infrared echellette (flat-fielding, wavelength calibration across all five orders, sky subtraction, telluric handling, flux calibration, plus the manual object extraction the paper itself had to use because the source-arc continuum is below the noise) is out of scope and is not reproducible to the ± 0.001 redshift target in a single session.

3. The paper tabulates no observed-frame line wavelengths. It reports only the final z_s and the per-system line *lists*, so a pure “ z_s from published λ_{obs} ” inversion is not available directly from the tables — which is why the arithmetic check runs the relation forward ($z_s \rightarrow \lambda_{\text{obs}}$) rather than backward.

What we therefore validate. Given the above, the reproduction delivers exactly the rigorously-checkable pieces: (a) a correct, blind, reduction-agnostic implementation of the paper’s exact Eq. 1 fit model — `fit_redshift()` takes any reduced (λ , flux, err) window and returns z_s — proven to recover all six published z_s to $\ll 0.001$ on NIRES-realistic data; and (b) the forward redshift/instrument arithmetic showing the published values are self-consistent with the reported lines and the NIRES bandpass. The KOA pipeline that would feed a real reduction is built and verified.

Not reproduced. The PYPEIT reduction itself; the absolute flux calibration and photometry; the lens redshifts z_d (taken from DESI, Lick Kast, and other sources, not re-derived here); the second lensed-source redshift within J006 and the DESI-secured source redshift for J023; and any lens modeling or Einstein-radius extraction (deferred to the Foundry I modeling reproduction).

From proxy to real measurement. The path is mechanical and needs no code change: download the located Level-0 frames (`02_koa_query.py` with `DOWNLOAD=1`; the manifest in `data/koa_frame_manifest.json` lists exactly which KOAIDs to fetch), reduce them with PYPEIT on an x86 box or container, then call `fit_redshift()` from `03_linefit.py` on the resulting reduced 1D spectra.

7 Reproducibility caveats

Synthetic, not observed, spectra. The Table 1 fits run on synthetic spectra built at the published z_s . They demonstrate that the fit model is correctly implemented and that the published redshifts are recoverable from data of realistic NIRES quality; they do *not* demonstrate that we would obtain the same z_s from the observatory’s actual photons, because that would require the reduction step in §6.

Blind-seed outliers. The fully-blind brightest-pixel seed occasionally (in $\lesssim 0.7\%$ of Monte-Carlo draws for two systems) locks onto a noise spike, producing a catastrophic Δz . In practice the paper’s human-in-the-loop line identification eliminates this failure mode; passing an approximate z_0 to `fit_redshift()` likewise removes it. We report the median σ_z (robust to these outliers) as the per-fit precision comparable to the paper.

Line lists and rest wavelengths. The per-system line lists, the chosen fit-line pairs, and the (z_d, z_s) values are transcribed by hand from the paper’s Tables 1–2; vacuum rest wavelengths are standard. Any transcription error would propagate directly into the consistency check.

8 Software and data availability

- Scripts: `reproductions/foundry-iii/01--05*.py`; this report builds against the artifacts they produce in `data/` (gitignored, regenerable) and `figs/`.
- Keck Observatory Archive (Berriman et al., 2022), queried via `pyKOA` (NASA Exoplanet Science Institute, 2023): <https://koa.ipac.caltech.edu>. NIRES serves Level-0 frames only.

- Python environment: `/home/benson/.venvs/lens` — `numpy` (Harris et al., 2020), `scipy` (Virtanen et al., 2020), `astropy` (Astropy Collaboration, 2022), `astroquery`, `pykoa`, `pymupdf`, `matplotlib`.
- PYPEIT (Prochaska et al., 2020a,b) is the paper’s reduction package; it does not build on the present aarch64 host (§6).
- Code repository: <https://github.com/usfcs-edu/agentlc-lensing>.
- Companion reproductions: Foundry I (reproductions/foundry-i/, Huang et al. 2025) and the Hsu+2025 pair-wise spectroscopic search (reproductions/hsu-2025/, Hsu et al. 2025).

Software: NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Astropy (Astropy Collaboration, 2022), pyKOA (NASA Exoplanet Science Institute, 2023), PypeIt (Prochaska et al., 2020a).

Facility: Keck:II (NIRES), Mayall (DESI), HST.

9 Conclusions

We reproduce the redshift-measurement core of Agarwal et al. (2025). A from-scratch implementation of the paper’s exact two-Gaussian, six-parameter fit model (Eq. 1) recovers all six published NIRES source redshifts ($z_s = 1.675\text{--}3.332$) blind to $|\Delta z| < 0.001$, with worst single-shot offset 2.2×10^{-4} and per-fit precision $\sigma_z \approx 0.9\text{--}1.4 \times 10^{-4}$ matching the paper’s quoted $O(10^{-4})\text{--}O(10^{-5})$. A closed-form redshift-arithmetic check confirms every reported line lands inside the NIRES $0.94\text{--}2.45\ \mu\text{m}$ band and that all six sources’ [O II] $\lambda 3727$ is redshifted past the DESI optical edge, reconstructing the paper’s near-infrared motivation. The KOA access pipeline locates all eight systems’ raw frames with per-system counts matching the published exposure times.

This is an honest consistency reproduction, not a measurement from the actual reduced spectra: KOA serves only raw NIRES data and PYPEIT does not build on the available hardware, so the from-raw-frames echelle reduction is out of scope. The fitter is reduction-agnostic and drops directly onto reduced 1D spectra once they exist — the only remaining step to convert this proxy into a genuine re-measurement.

A Side-by-side comparison with the published figures and tables

For visual verification, each block below pairs a table reproduced in this report (**a, top**) with the corresponding table from the published paper (**b, bottom**). Published grids are reproduced from Agarwal et al. (2025) for comparison. Headline agreement is restated in each caption; the full discussion is in §3 and §6.

References

- S. Agarwal, X. Huang, W. Sheu, C. Storfer, M. Tamargo-Arizmendi, S. Tabares-Tarquinio, D. J. Schlegel, et al. DESI Strong Lens Foundry III: Keck Spectroscopy for Strong Lenses Discovered Using Residual Neural Networks. *arXiv preprint*, 2025. Submitted to ApJ.
- Astropy Collaboration. The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935:167, 2022. doi: 10.3847/1538-4357/ac7c74.

(a) This work (Table 1)

System	fit lines	z_s^{pub}	z_s^{fit}	Δz	σ_z
DESI J006.3643+10.1853	H α + [O III]5007	2.39688	2.39700	$+1.2 \times 10^{-4}$	1.4×10^{-4}
DESI J094.5639+50.3059	[O III]5007 + H β	3.33185	3.33163	-2.2×10^{-4}	1.6×10^{-4}
DESI J133.3800+23.3652	H α + [O III]5007	2.18858	2.18863	$+4.7 \times 10^{-5}$	1.1×10^{-4}
DESI J154.5307-00.1368	H α + H β	1.73810	1.73816	$+6.1 \times 10^{-5}$	6.3×10^{-5}
DESI J165.4754-06.0423	H α + [O III]5007	1.67511	1.67523	$+1.2 \times 10^{-4}$	7.5×10^{-5}
DESI J215.2654+00.3719	H α + [O III]5007	2.20645	2.20638	-7.3×10^{-5}	1.4×10^{-4}
6/6 within ± 0.001; worst $\Delta z = 2.2 \times 10^{-4}$.					

(b) Agarwal et al. (2025), Table 2

System Name	z_d	z_s	σ_{z_s}
DESI J006.3643+10.1853	0.4631 ^a	2.39688	± 0.00004
		1.3143 ^a	$\pm 0.00006^a$
DESI J023.0157-16.0040	-	1.5818 ^a	$\pm 0.00007^a$
DESI J024.1631+00.1384	0.3445 ^a	-	-
DESI J094.5639+50.3059	0.522 ^b	3.33185	± 0.00010
DESI J133.3800+23.3652	0.3053 ^a	2.18858	± 0.00002
DESI J154.5307-00.1368	0.3718 ^a	1.73735, 1.73885	$\pm 0.00004, 0.00006$
DESI J165.4754-06.0423	0.4834 ^a	1.67511	± 0.00005
DESI J215.2654+00.3719	0.6566 ^a	2.20645	± 0.00035

Figure 2: Source redshifts. (a) This work: our blind fit of the paper’s exact two-Gaussian model (Eq. 1) to NIRES-realistic synthetic spectra, listing the published z_s^{pub} against our recovered z_s^{fit} . (b) The published redshift results; the bold z_s column is the counterpart of our z_s^{pub} . All six NIRES source redshifts ($z_s = 1.675\text{--}3.332$) are recovered to $|\Delta z| < 0.001$, worst single-shot 2.2×10^{-4} . Panel (b) reproduced from Agarwal et al. (2025), their Table 2, for comparison.

G. B. Berriman, C. R. Gelino, A. C. Laity, et al. The Keck Observatory Archive (KOA). <https://koa.ipac.caltech.edu>, 2022. NASA Exoplanet Science Institute / W. M. Keck Observatory.

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(a) This work (Table 3)

System	UT night	N_{frame}	t_{exp} (s)
DESI J006.3643+10.1853	2022-11-15	4	1200
DESI J023.0157−16.0040	2022-11-15	2	600
DESI J024.1631+00.1384	2022-11-15	0	—
DESI J094.5639+50.3059	2022-11-15	12	3600
DESI J133.3800+23.3652	2023-01-11	16	4800
DESI J154.5307−00.1368	2022-11-15	8	2400
DESI J165.4754−06.0423	2023-01-11	12	3600
DESI J215.2654+00.3719	2023-01-11	4	1200

(b) Agarwal et al. (2025), Table 1

Date	Seeing	Airmass	Target	RA	Dec	Isophotal Magnitude	Contour Area	Exp. Time (s)
Nov 13, 2022	0.55''	1.38	DESI J006.3643+10.1853	6.3643	+10.1853	23.84 ± 0.02	0.0325	1200
						23.87 ± 0.02	0.0450	
		1.60	DESI J023.0157-16.0040	23.0157	−16.0040	24.02 ± 0.02*	0.1014	600
						23.23 ± 0.02*	0.1436	
		1.56	DESI J024.1631+00.1384	24.1631	+0.1384	22.78 ± 0.01	0.1925	600
						22.71 ± 0.01	0.2425	
		1.18	DESI J094.5639+50.3059	94.5639	+50.3059	23.85 ± 0.02	0.0775	3600
						24.36 ± 0.03	0.0325	
		1.43	DESI J154.5307-00.1368	154.5307	-0.1368	24.24 ± 0.03*	0.1050	2400
						25.03 ± 0.04	0.0350	
						23.51 ± 0.02	0.0350	2400
						23.34 ± 0.02	0.0525	
Jan 10, 2023	0.8''	1.17	DESI J165.4754-06.0423	165.4754	−6.0423	22.15 ± 0.01	0.4775	3600
						22.28 ± 0.01	0.3725	
		1.11	DESI J215.2654+00.3719	215.2654	+0.3719	24.15 ± 0.03	0.0925	2400
						24.41 ± 0.03	0.0800	

Figure 3: Keck nires observation log. (a) This work: the per-system raw NIRES science frames located in the Keck Observatory Archive, with the recovered $N_{\text{frame}} \times 300\text{s}$ on-source exposure time. (b) The published observation log; the rightmost *Exp. Time (s)* column is the counterpart of our t_{exp} . The KOA query recovers all eight systems’ nights and sky positions, and five of the eight tabulated on-source exposure times are reproduced exactly (J006, J023, J094, J154, J165). J024 returns zero positionally-matched science frames (a non-detection), while the archive yields more matched frames than the paper’s nominal value for J133 (4800 vs 2400s) and fewer for J215 (1200 vs 2400s). Panel (b) reproduced from Agarwal et al. (2025), their Table 1, for comparison.

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